Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of **Smartphone Usage & Addiction**

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Introduction:

Smartphones have quietly become one of the most dominant forces in modern daily life — not just as communication tools, but as constant companions that shape how we sleep, study, and manage stress. While this shift has brought undeniable convenience, it has also raised serious questions about where healthy usage ends and addiction begins.
This report examines smartphone usage patterns across 7,500 users, exploring the behavioral and demographic factors most associated with addiction. Rather than simply describing the data, we aim to test specific hypotheses and draw meaningful conclusions — ones that could be relevant to students, mental health professionals, and anyone trying to understand the role smartphones play in our wellbeing.

Research Question:

This study aims to answer the following research question:

"To what extent does daily smartphone screen time contribute to addiction, and how do behavioral and demographic factors such as sleep hours, stress level, and gender relate to smartphone dependency among users?"

Hypothesis:

Hypothesis 1 — Daily Screen Time & Addiction (T-Test)
H_0 (Null): There is no difference in mean daily screen time between addicted and non-addicted individuals.
H_1 (Alternative): There is a difference in mean daily screen time between addicted and non-addicted individuals.

Hypothesis 2 — Screen Time & Sleep (Pearson Correlation)
H_0 (Null): There is no linear relationship between screen time and sleep hours.
H_1 (Alternative): There is a significant linear relationship between screen time and sleep hours.

Hypothesis 3 — Gender & Addiction (Chi-Square)
H_0 (Null): Gender and addiction are independent — there is no association between them.
H_1 (Alternative): Gender and addiction are significantly associated.

Hypothesis 4 — Stress Level & Addiction (Chi-Square)
H_0 (Null): Stress level and addiction are independent.
H_1 (Alternative): Stress level and addiction are significantly associated.

Population of Interest:

The population of interest is smartphone users between the ages of 18 and 35 years old, regardless of gender or geographic location. This age group was selected as it represents the demographic most likely to exhibit heavy smartphone usage and dependency behaviors, being the primary consumers of social media, gaming, and digital communication platforms.

Sampling Method:

The dataset was collected using a convenience sampling approach, gathering data from 7,500 smartphone users across varying demographics including age, gender, and stress levels. Although convenience sampling may not perfectly represent the global population, it remains the most practical approach for this scale of study. Steps were taken to ensure demographic balance gender and stress level are nearly equally distributed across the dataset which minimizes sampling bias and strengthens the reliability of the analysis.

Bias Identification:

The dataset used in this analysis may contain several sources of bias that could affect the validity and generalizability of the results. First, sampling bias may be present if the data overrepresents certain groups, such as younger or more tech-savvy individuals, while underrepresenting others like older populations. Gender representation may also be imbalanced, potentially skewing conclusions related to differences in addiction behavior. Additionally, many variables, such as daily screen time, sleep hours, and stress level, are likely self-reported, introducing inaccuracies due to overestimation, underestimation, or social desirability bias. The labeling of addiction (e.g., categorizing users as addicted or not) may involve subjective thresholds, leading to possible misclassification, especially for borderline cases. There is also a risk of feature correlation bias, as multiple variables related to screen usage may overlap and disproportionately influence results. Missing data in some fields could further distort findings if not handled properly. Moreover, the dataset lacks important contextual factors such as mental health or socioeconomic status, which limits deeper interpretation, and since the data appears to be cross-sectional, it does not capture changes over time. Finally, some variables are measured in a simplified or subjective way, reducing precision. Overall, these biases should be considered when interpreting the analysis, and appropriate preprocessing and validation techniques are necessary to minimize their impact.

Survey Questions/Collected Data/Dataset:

Overview

The dataset used in this analysis is the Smartphone Usage and Addiction Analysis dataset, containing 7,500 records and 14 features. It was sourced from Kaggle and covers a range of behavioral, demographic, and usage-related variables.

Feature Descriptions

Feature	Type	Desctreption
Age	numerical	Ages of the users (18-35)
Gender	categorical	Male-Female-other
daily_screen_time_hours	numerical	Average daily screen usage in hours
social_media_hours	numerical	Hours spen on social media per day
gaming_hours	numerical	Hours spent gaming per day
work_study_hours	numerical	Hours used for study or work per day
sleep_hours	numerical	Average sleep hours per night
notifications_per_day	numerical	Number of notifications received per day
app_opens_per_day	numerical	Number of times apps opend per day
weekend_screen_time	numerical	Screen time specifically on weekends
stress_level	categorical	Perceived stress: Low, Medium, High
academic_work_impact	categorical	Whether smartphone impacts academic work (Yes/No)

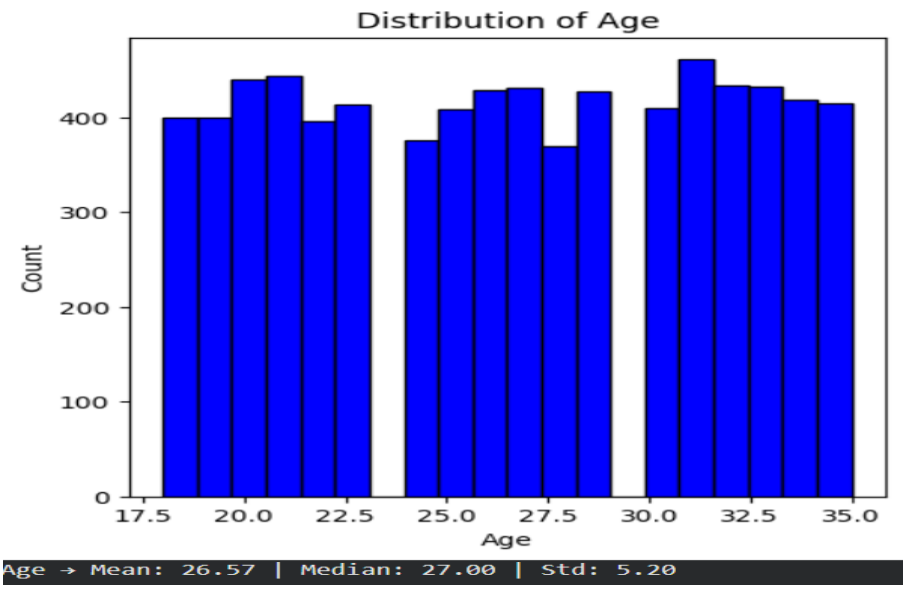
addiction_level	categorical	Self-reported addiction: Mild, Moderate, Severe
addicted_label	categorical	Target variable: 0 = Not Addicted, 1 = Addicted

Number of samples used: 7500.

Analysis:

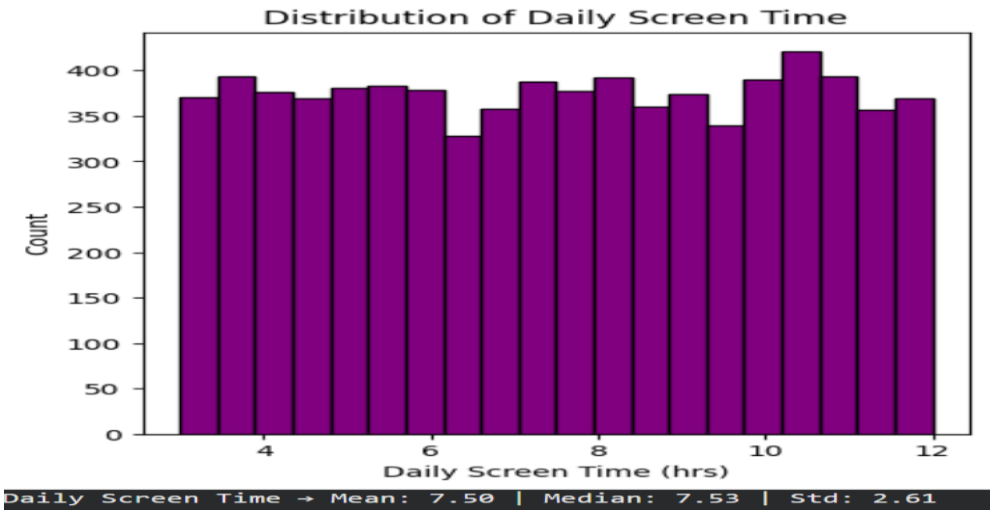
To analyze the dataset, we calculated descriptive statistics such as the mean, median, and standard deviation for key numerical variables. We also created multiple visualizations to better understand the data distribution and identify patterns related to smartphone usage and addiction.

1. **Distribution of Age:**



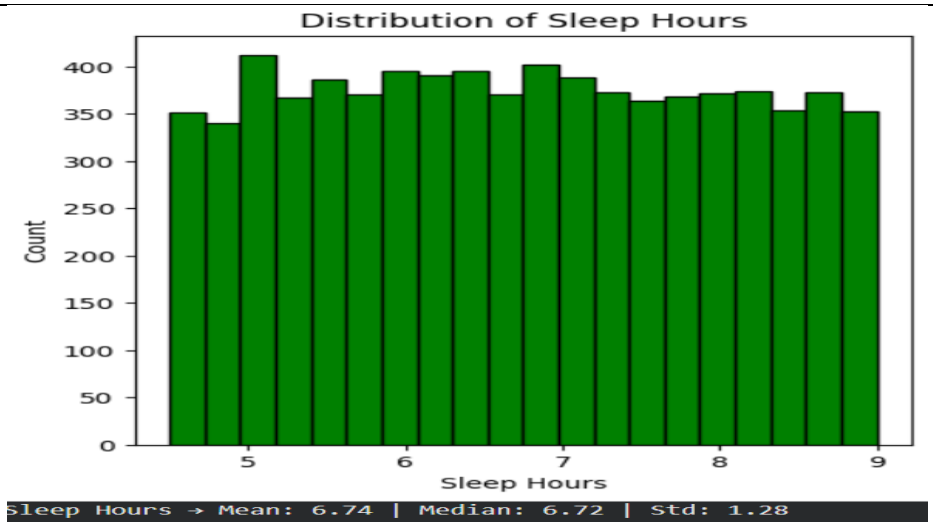
This histogram represents the age distribution of the study's population, which primarily consists of young adults between the ages of **18 and 35**. The data shows an approximately uniform and normal distribution, with a **Mean** age of **26.57** and a **Median** of **27.00**. The small difference between the mean and median, along with a **Standard Deviation** of **5.20**, indicates that the dataset is well-balanced across this specific age range. This ensures that the findings regarding smartphone addiction are representative of the target demographic most active on digital platforms.

2. Distribution of Daily Screen Time



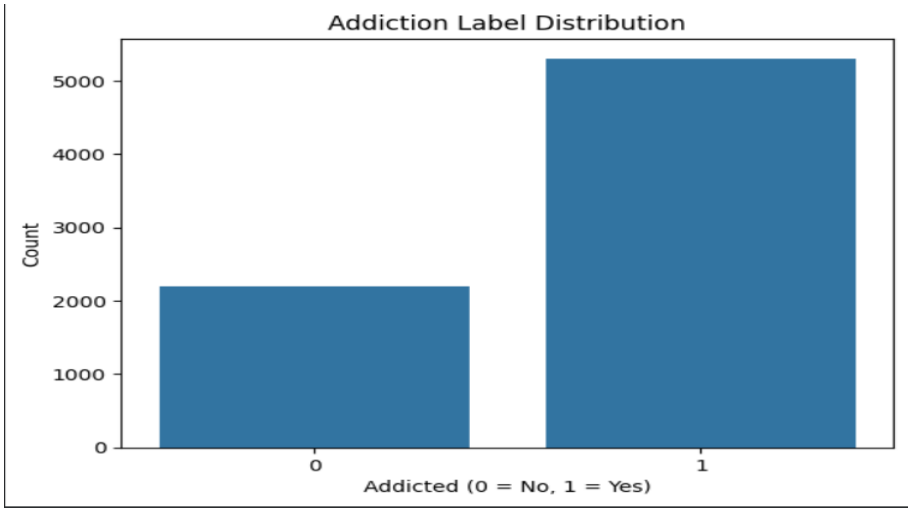
The histogram illustrates the distribution of daily screen time among the 7,500 participants. The data shows that screen usage is consistently high across the sample, with an average (Mean) of 7.50 hours and a Median of 7.53 hours. The usage levels are relatively evenly distributed between 3 and 12 hours, though a notable peak is observed in the 10+ hours category. This widespread high usage is identified as a primary factor linked to both smartphone addiction and potential sleep deprivation within this study.

3. Distribution of Sleep Hours



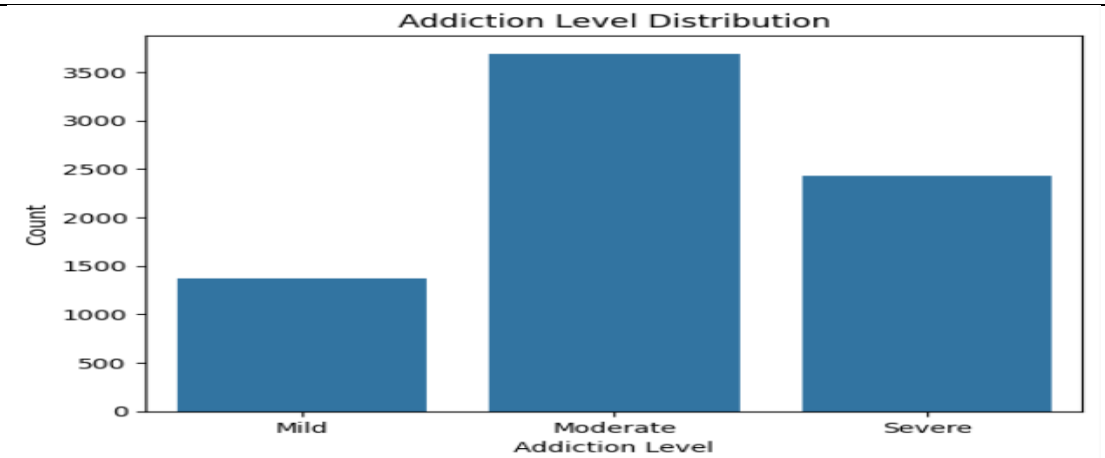
The histogram illustrates the distribution of sleep hours among the participants, with values ranging from 4.5 to 9 hours. The dataset shows a consistent and nearly uniform distribution of sleep durations, with a Mean of 6.74 hours and a Median of 6.72 hours. With a Standard Deviation of 1.28, the data indicates that while most users maintain moderate sleep patterns, a significant number of individuals fall below the recommended sleep duration. This distribution is crucial for analyzing how smartphone usage patterns might be encroaching on the rest periods of users within the sample.

4. Addiction Label Distribution



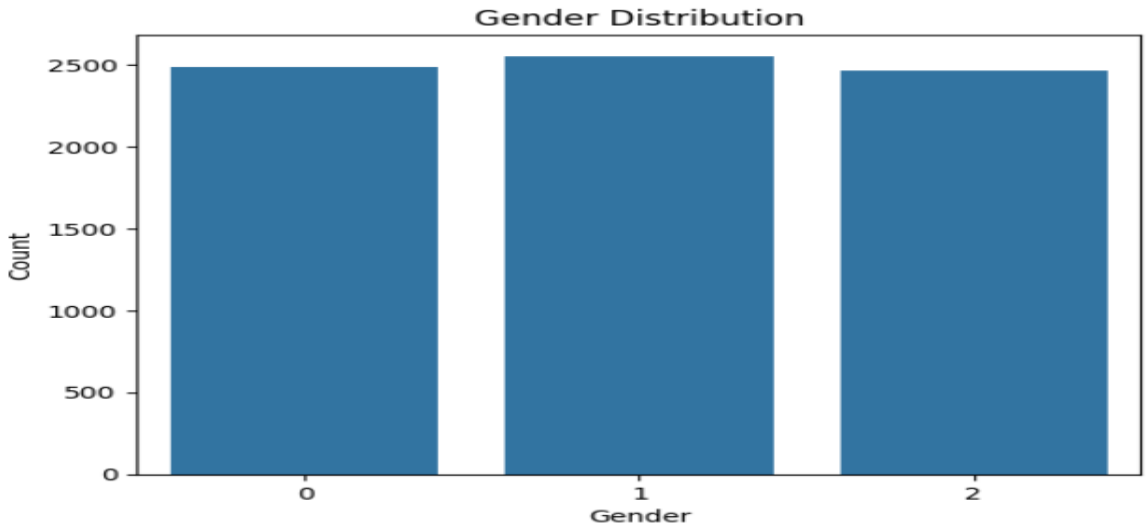
This count plot illustrates the binary classification of users within the dataset based on their addiction status. The chart shows a significant disparity between the two groups, with the majority of the **7,500** participants being classified as addicted (**Label 1**). Specifically, more than **5,000** users are identified as addicted, while approximately **2,200** users are labeled as non-addicted (**Label 0**). This high prevalence of addiction within the sample underscores the severity of smartphone dependency among the studied demographic and provides a substantial baseline for the subsequent predictive modeling.

5. Addiction Level Distribution



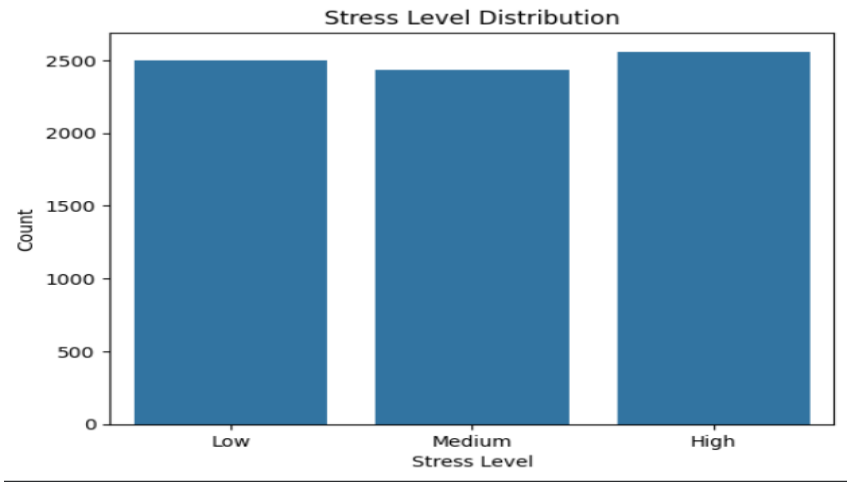
This visualization breaks down the addiction categories among the surveyed participants into three primary levels: Mild, Moderate, and Severe. The chart indicates that the Moderate level is the most prevalent, with over 3,500 individuals falling into this category. The Severe addiction level follows with approximately 2,400 users, while the Mild category represents the smallest portion of the sample at around 1,400 users. These results highlight that while extreme addiction cases are significant, the majority of users experience a moderate degree of smartphone dependency, which still poses a substantial impact on their daily lives.

6. Gender Distribution



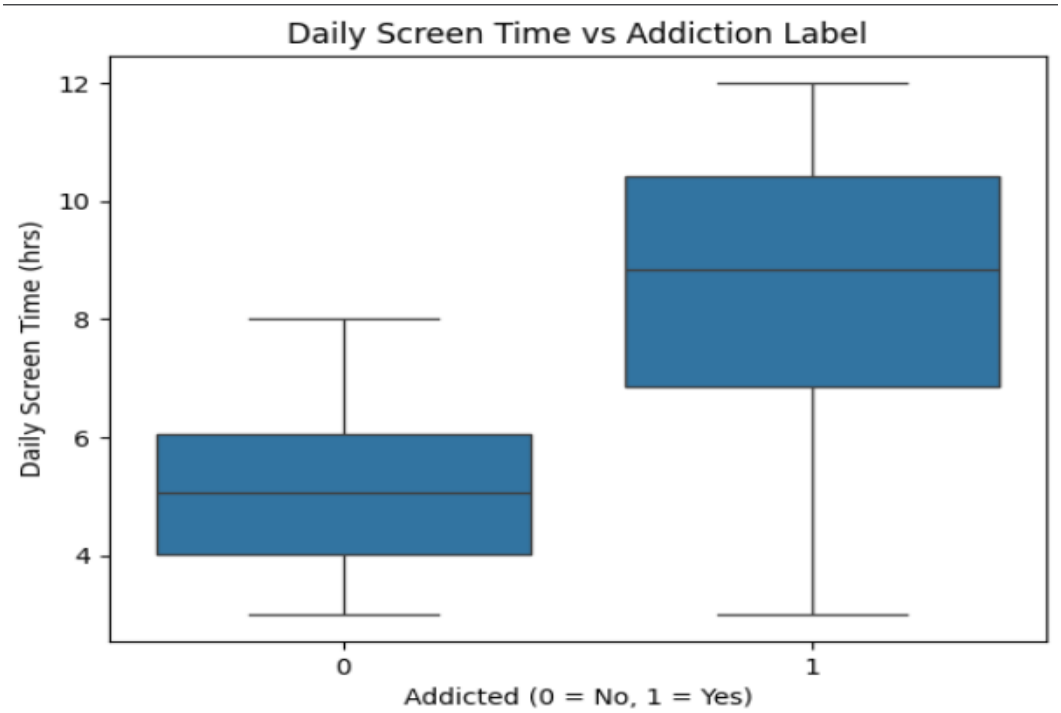
This count plot illustrates the distribution of participants by gender within the dataset, where the categories are encoded as **0 (Other)**, **1 (Male)**, and **2 (Female)**. The chart reveals a highly balanced representation across all three groups, with each category containing approximately **2,500** individuals. This near-equal distribution is a significant strength of the dataset, as it minimizes gender bias and ensures that the subsequent analysis of smartphone addiction is equally applicable and fair across different gender identities.

7. Stress Level Distribution



This count plot displays the distribution of perceived stress levels among the **7,500** participants, categorized into **Low**, **Medium**, and **High** groups. The visualization shows a very balanced representation across all three categories, with each stress level containing approximately **2,500** individuals. This balance is critical for the study as it allows for an unbiased evaluation of how different levels of psychological stress relate to smartphone addiction behaviors without one group over-shadowing the others.

8. Daily Screen Time vs Addiction

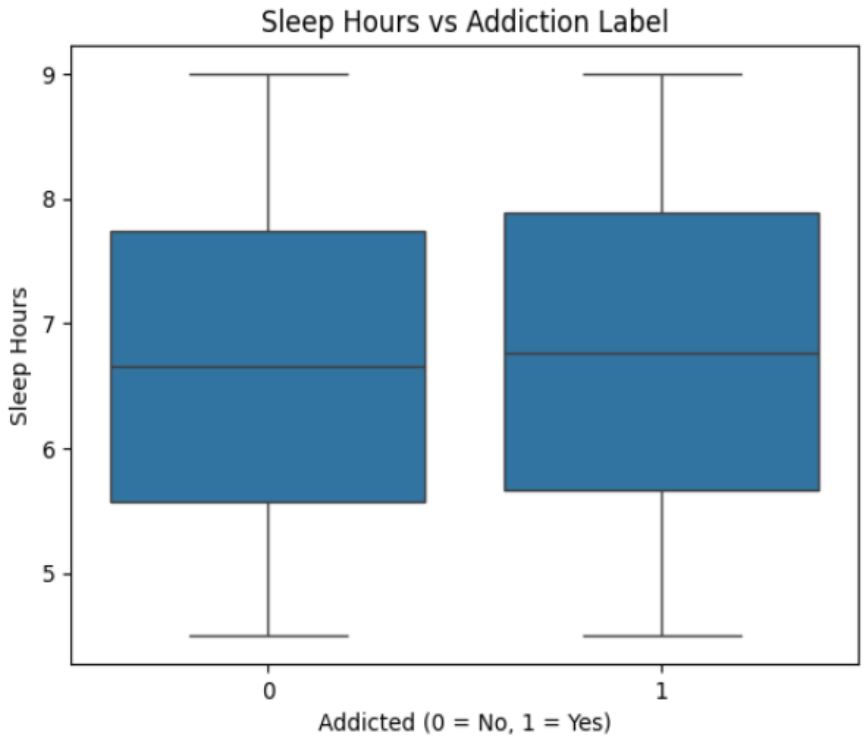


This boxplot compares the daily screen time (in hours) between non-addicted (0) and addicted (1) users. The visualization reveals a clear and significant difference between the two groups:

- Addicted users (1) generally have much higher screen time, with a median of approximately 9 hours, and their usage reaches up to the maximum of 12 hours.
- Non-addicted users (0) show a significantly lower median of around 5 hours, with most of their usage falling below 6 hours.

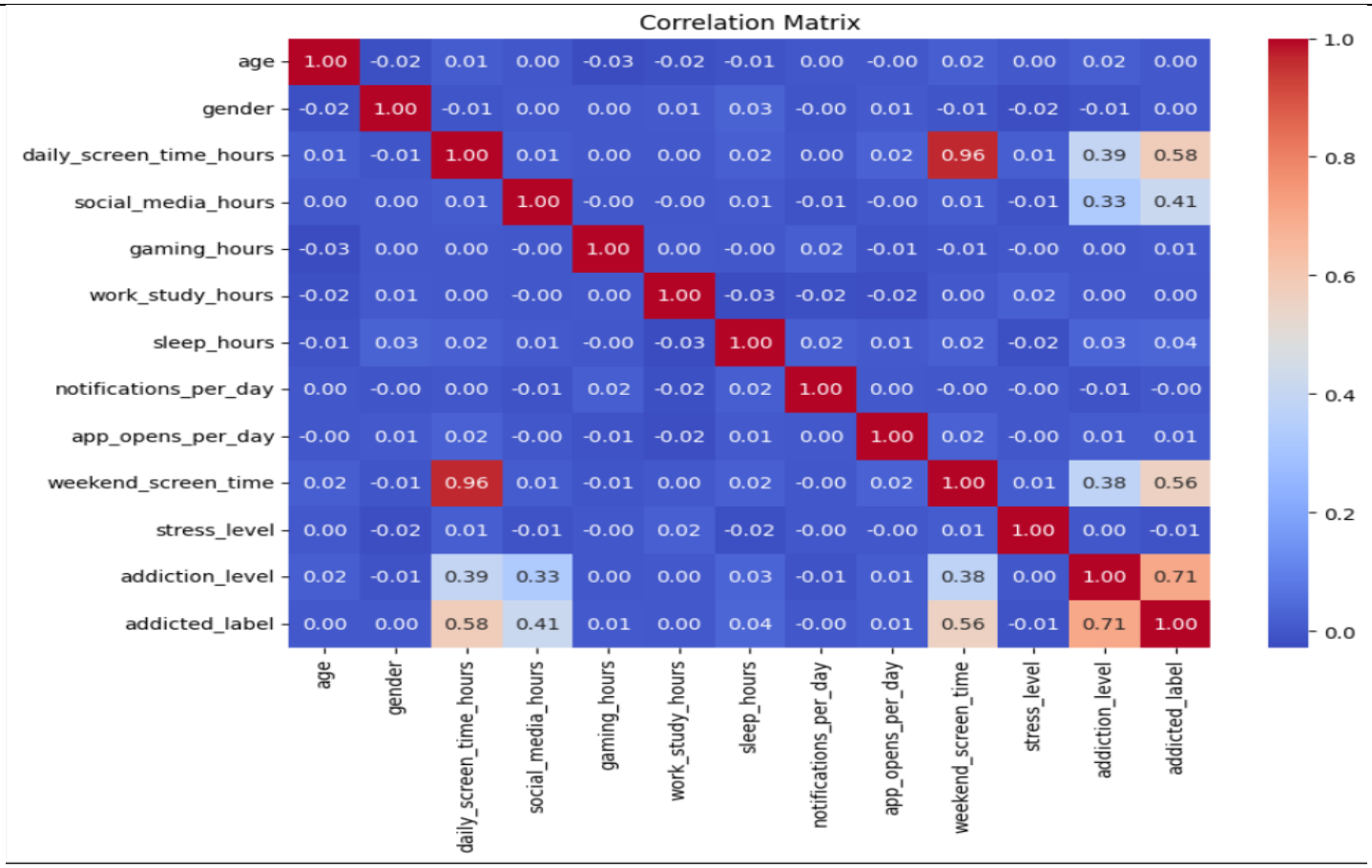
The lack of overlap between the interquartile ranges (the boxes) of the two groups strongly suggests that daily screen time is a major predictor of smartphone addiction in this dataset. This visual evidence aligns with the earlier statistical findings and provides a clear justification for why screen time was identified as the most important feature in the machine learning models.

9. Sleep Hours vs Addiction



This boxplot compares the distribution of average sleep hours per night between non-addicted (0) and addicted (1) individuals. Surprisingly, the visualization shows that both groups have almost identical sleep patterns, with a median of approximately 6.7 hours for both addicted and non-addicted users. The interquartile ranges (the boxes) and the minimum/maximum values are largely overlapping, indicating that there is no substantial difference in sleep duration between the two groups within this dataset. This visual finding directly supports the results of the Pearson Correlation and Hypothesis Testing performed earlier, which failed to reject the null hypothesis, confirming that sleep hours are not a significant indicator of smartphone addiction in this specific study.

10. Correlation Matrix

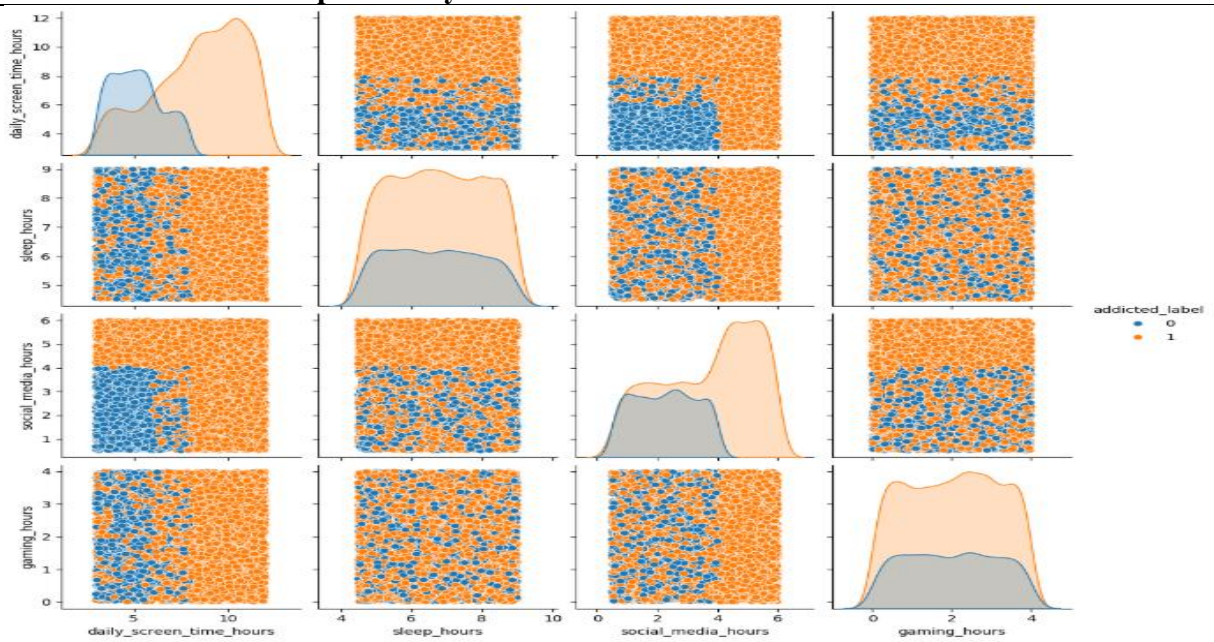


This heatmap provides a comprehensive overview of the linear relationships between all numerical variables in the dataset. The color intensity and the numbers (ranging from -1.00 to 1.00) indicate the strength and direction of these correlations. Key insights from the matrix include:

- **Strong Positive Correlations**: A very high correlation of **0.96** exists between daily_screen_time_hours and weekend_screen_time, indicating that usage habits are consistent throughout the week.
- **addiction Drivers**: The target variable addicted_label shows its strongest positive correlations with addiction_level (**0.71**), daily_screen_time_hours (**0.58**), and weekend_screen_time (**0.56**). This reinforces the finding that total screen time is the most significant factor associated with addiction.
- **Weak Relationships**: Variables such as age, gender, and stress_level show near-zero correlation coefficients with the addiction label, confirming that they do not have a strong linear relationship with smartphone dependency in this sample.
- **Social Media Impact**: social_media_hours also shows a moderate positive correlation with addiction (**0.41**), suggesting it is a major component of addictive screen behavior.

Overall, the heatmap justifies the selection of screen-time-related features for the machine learning models, as they demonstrate the most meaningful statistical connections to the target variable.

11. Pairplot Analysis



This pairplot visualizes the pairwise relationships between four key numerical variables: daily_screen_time_hours, sleep_hours, social_media_hours, and gaming_hours, with the data points colored by their addicted_label (0: Blue, 1: Orange).

- Clear Separation in Screen Time: The diagonal density plots (top-left) clearly show two distinct peaks for daily_screen_time_hours. Non-addicted users (blue) are clustered at lower hours, while addicted users (orange) are heavily concentrated at higher usage levels.

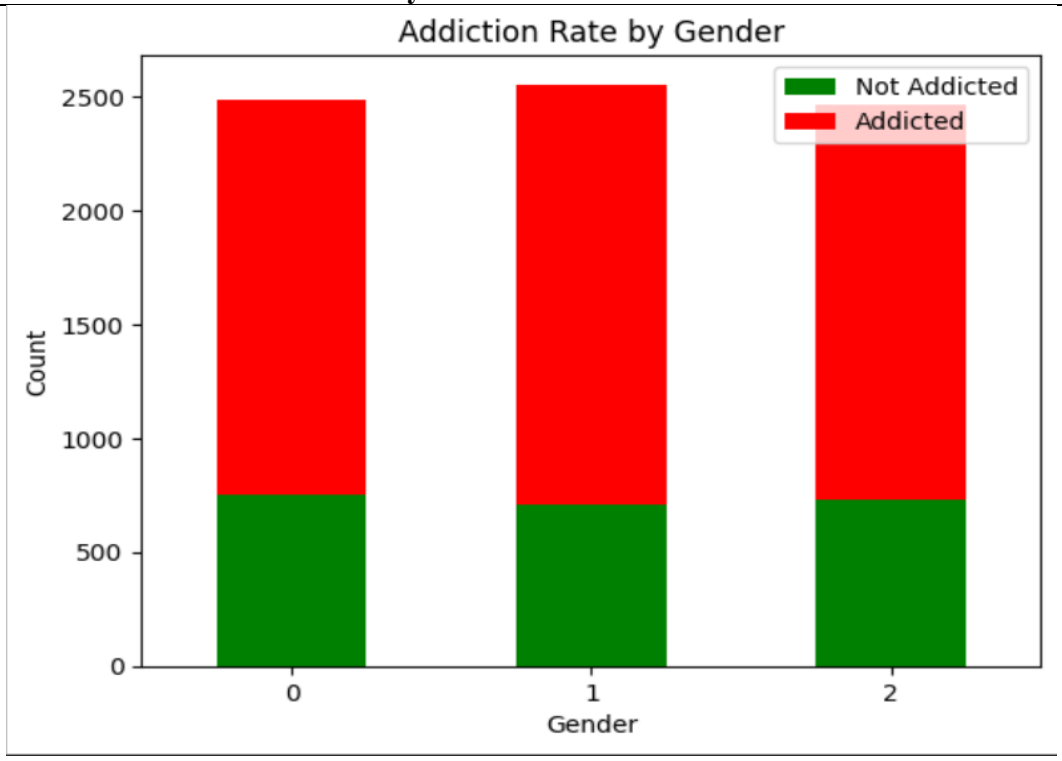
- Social Media & Addiction: Similar to screen time, the density plot for social_media_hours shows a significant shift, where addicted individuals (orange) spend considerably more time on social media compared to non-addicted individuals.

- Overlapping Sleep & Gaming Patterns: In contrast, the density plots for sleep_hours and gaming_hours show significant overlap between both groups. This indicates that these factors, on their own, do not distinguish between addicted and non-addicted users as clearly as total screen time or social media usage does.

- Scatter Plot Insights: The scatter plots (off-diagonal) highlight that most addicted users (orange points) occupy the upper ranges of usage hours across all screen-related activities, creating a clear "cluster" of dependency behavior.

This multi-dimensional view confirms that while some behaviors like gaming or sleep are distributed similarly across the population, high screen time and high social media consumption are the primary signatures of smartphone addiction in this dataset.

12. Addiction Rate by Gender



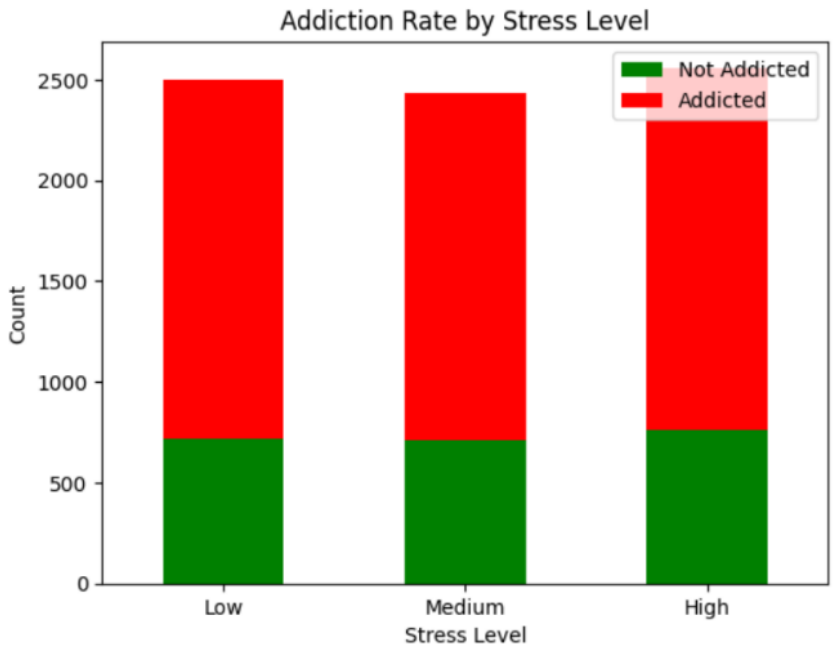
This stacked bar chart illustrates the proportion of Addicted (Red) versus Not Addicted (Green) individuals within each gender category (0: Other, 1: Male, 2: Female).

- **Uniform Distribution:** The visualization shows that the ratio of addicted to non-addicted users is remarkably consistent across all genders. In every category, the vast majority of participants are classified as "Addicted," with the red segments significantly outsize the green segments.

- **Demographic Neutrality:** The fact that approximately 70-75% of each gender group is addicted suggests that gender is not a determining factor for smartphone addiction in this dataset.

- **Data Balance:** This graph reinforces the reliability of the study, as it demonstrates that the high addiction rates observed are a general trend in the sample rather than being skewed toward a specific gender group.

13. Addiction Rate by Stress Level



This stacked bar chart visualizes the relationship between perceived stress levels (Low, Medium, High) and the prevalence of smartphone addiction (Red: Addicted, Green: Not Addicted).

- **Consistent Addiction Levels:** Similar to the gender distribution, the addiction rate remains remarkably stable across all three stress categories. Whether an individual reports low or high stress, the proportion of addiction stays around the same majority level.
- **Independence of Features:** This visualization confirms that in this specific dataset, stress level is not a primary driver or a prerequisite for smartphone addiction. Individuals across the entire stress spectrum show similar patterns of dependency.
- **Statistical Alignment:** These findings provide visual context for the low correlation coefficient observed in the correlation matrix between stress_level and addicted_label. It suggests that addiction behaviors are likely more tied to usage habits (like screen time) than to the user's psychological stress state.

The analysis indicates that **daily screen time** is the most **important** factor associated with smartphone addiction. In contrast, variables such as **gender** and **stress level** show **weaker** relationships, suggesting that behavioral usage patterns are more influential than demographic or psychological factors within this dataset.

Hypothesis Testing Steps

Hypothesis 1 — T-Test: Screen Time vs Addiction

Step	Detail
Step 1: Hypotheses	H_0: No difference in mean screen time | H_1: Significant difference exists
Step 2: Test Used	Independent samples T-Test (ttest_ind)
Step 3: Results	t-statistic: 61.19 | p-value: 0.0
Step 4: Decision	$p < 0.05 \rightarrow$ Reject H_0

Conclusion: The t-test produced a highly significant result, confirming that addicted users spend considerably more time on their screens daily. Among all variables tested, screen time showed the strongest and most direct link to addiction status.

Hypothesis 2 — Pearson Correlation: Screen Time vs Sleep Hours

Step	Detail
Step 1: Hypotheses	H_0: No linear relationship between screen time and sleep | H_1: A linear relationship exists
Step 2: Test Used	Pearson Correlation (pearsonr)
Step 3: Results	$r = 0.019$ | p-value: 0.094
Step 4: Decision	$p > 0.05 \rightarrow$ Fail to reject H_0

Conclusion: With a near-zero correlation coefficient, the results suggest that higher screen time does not translate to fewer hours of sleep in this sample. Any relationship that may exist is too weak to be detected through a linear measure alone.

Hypothesis 3 — Chi-Square Test: Gender vs Addiction

Step	Detail
Step 1: Hypotheses	H_0: Gender and addiction are independent | H_1: Gender and addiction are associated
Step 2: Test Used	Chi-Square Test of Independence (chi2_contingency)
Step 3: Results	$\chi^2 = 2.382$ | p-value: 0.123
Step 4: Decision	$p > 0.05 \rightarrow$ Fail to reject H_0

Conclusion: Gender does not appear to be a determining factor in whether a user is classified as addicted. The observed differences in addiction rates between male and female respondents were small enough to be attributed to chance.

Hypothesis 4 — Chi-Square Test: Stress Level vs Addiction

Step	Detail
Step 1: Hypotheses	H_0: Stress level and addiction are independent | H_1: Stress level and addiction are associated
Step 2: Test Used	Chi-Square Test of Independence (chi2_contingency)
Step 3: Results	$\chi^2 = 2.498$ | p-value: 0.645
Step 4: Decision	$p > 0.05 \rightarrow$ Fail to reject H_0

Conclusion: Despite the commonly assumed link between stress and addictive behaviour, this test found no meaningful association between the two. The notably high p-value suggests that stress level, as measured in this dataset, has little to no bearing on addiction classification.

Machine Learning Modeling

Objective:

In addition to statistical analysis, machine learning techniques were applied to predict smartphone addiction based on user behavior and demographic features. This step aims to evaluate whether addiction status can be accurately predicted and to identify the most important factors contributing to addiction.

Data Preparation:

Before building the models, the dataset was prepared to be suitable for machine learning algorithms. Categorical variables were converted into numerical format using encoding techniques. The target variable (addicted_label) was separated from the input features.

The dataset was then split into two parts:

- Training set (80%) used to train the models
- Testing set (20%) used to evaluate performance

This approach ensures that the models are tested on unseen data, improving the reliability of the results.

Models Used:

Three classification models were implemented and compared:

- **Logistic Regression:** A simple and interpretable model used as a baseline for classification.
- **Decision Tree:** A model that makes decisions by splitting data into branches based on feature values.
- **Random Forest:** An advanced ensemble model that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Model Evaluation:

The models were evaluated using accuracy score, which measures the proportion of correct predictions.

The results showed that the Random Forest model achieved the highest accuracy compared to the other models. This indicates that it was more effective in capturing patterns within the dataset.

This also suggests that the relationship between variables and smartphone addiction is complex and better modeled using ensemble techniques rather than simple linear models.

Confusion Matrix:

A confusion matrix was used to evaluate the performance of the Random Forest model in more detail.

It provides a breakdown of:

- Correct predictions for addicted and non-addicted users
- Incorrect predictions (false positives and false negatives)

This helps in understanding how well the model distinguishes between different classes beyond overall accuracy.

Feature Importance:

Feature importance analysis was conducted using the Random Forest model to identify the most influential variables in predicting addiction.

The results showed that daily screen time is the most important feature, followed by variables such as social media usage and app activity.

This finding supports the earlier statistical analysis, confirming that screen time plays a major role in smartphone addiction.

Conclusion

The hypothesis testing revealed that screen time was the only variable with a statistically significant relationship to addiction ($t = 61.19$, $p < 0.05$). In contrast, no significant associations were found between addiction and sleep hours, gender, or stress level. These findings indicate that, within this dataset, screen time is the strongest behavioral indicator of smartphone addiction, while demographic and psychological variables did not demonstrate statistically significant effects.

The machine learning results further supported the statistical findings. The implemented models were able to predict addiction status with good accuracy, with the Random Forest model achieving the best performance. In addition, feature importance analysis identified daily screen time as the most influential variable in predicting addiction. This consistency between statistical analysis and predictive modeling enhances the overall reliability and robustness of the study.

However, these results should be interpreted with caution. Limitations related to data collection methods, variable design, and potential biases may have influenced the outcomes. Furthermore, while the machine learning models demonstrated strong predictive capabilities, they identify patterns rather than causal relationships. Therefore, the findings should be viewed as indicative of associations rather than definitive evidence of causation.

Any potential issues

Some limitations should be considered. Categorical variables were encoded into numbers, which may affect interpretation. Correlation was applied to encoded data, which may not accurately represent real relationships. The analysis shows associations rather than causation. In addition, outliers and possible sample bias may influence the results.